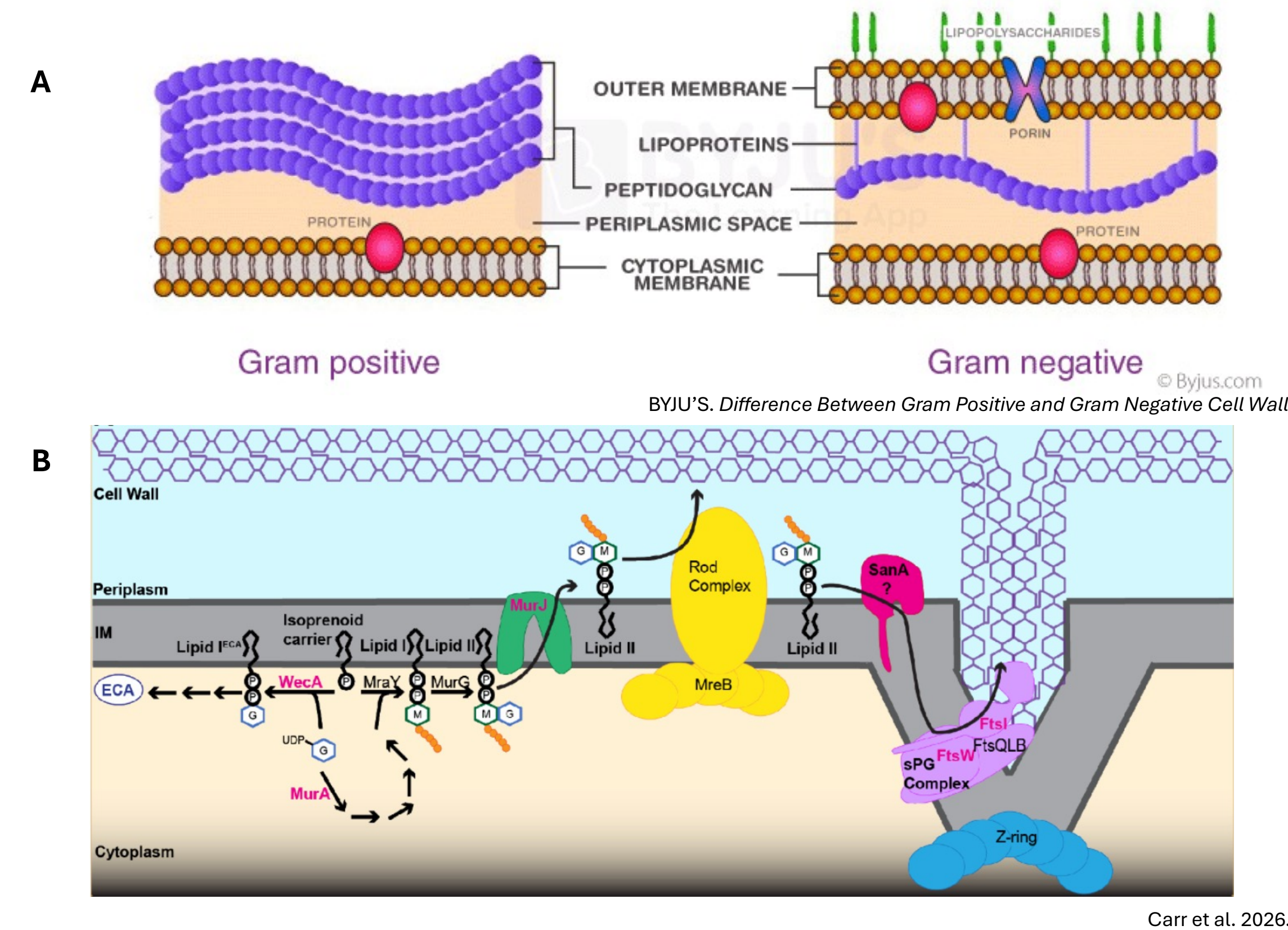


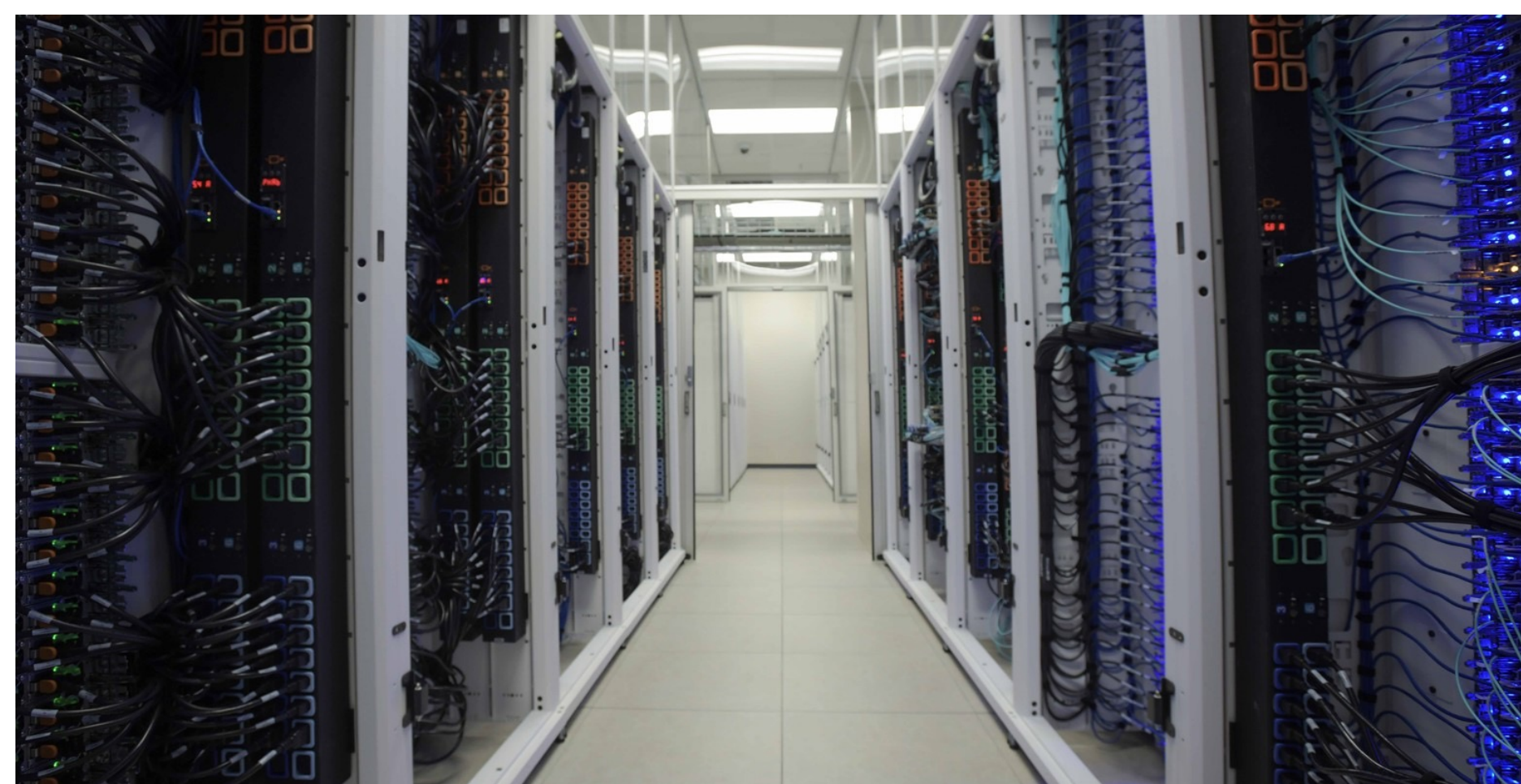
Introduction

- Escherichia coli* is a Gram-negative bacterium with a multilayered cell envelope that includes an outer membrane and a thin peptidoglycan layer.
- Stability of the peptidoglycan layer is essential for maintaining envelope integrity and resisting antibiotics and detergents.
- SanA is implicated in preserving cell wall homeostasis during cell division.
- The molecular mechanism of SanA function and its interaction partners remain unknown.



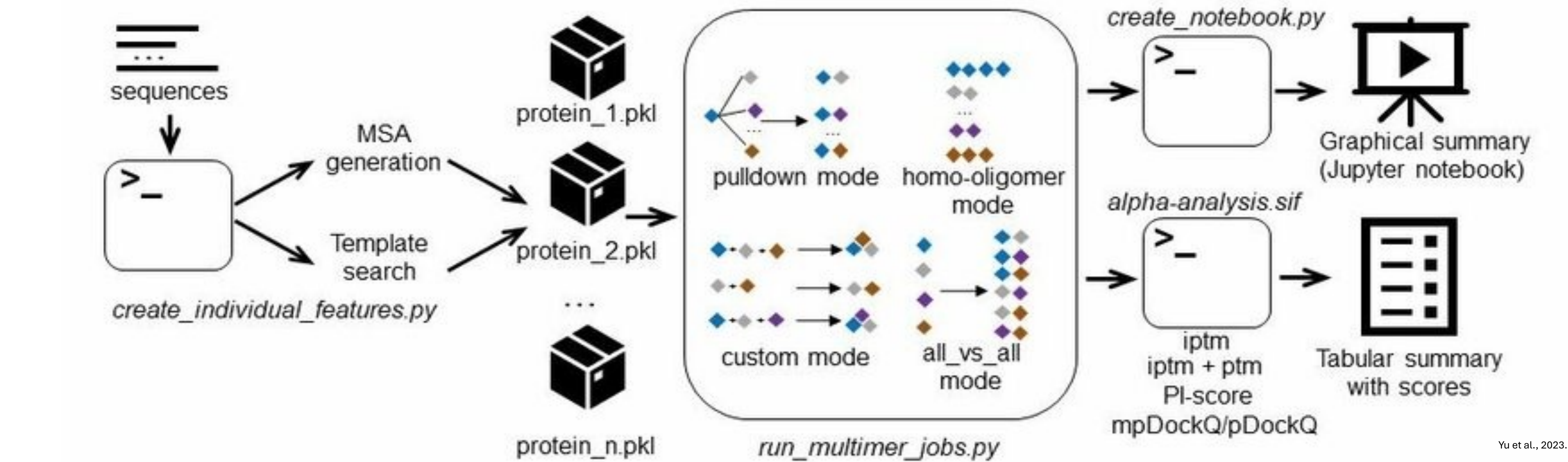
(A) Comparison of Gram-positive and Gram-negative cell envelope architecture. (B) *E. coli* inner membrane near the Z-ring, illustrating SanA's proposed role in regulating Lipid II levels and FtsI/FtsW's role in peptidoglycan synthesis.

Texas A&M HPRC Grace: High-Performance Computing for Large-Scale Biological Modeling



System Name:	Grace	https://hprc.tamu.edu/kb/User-Guides/Grace/Hardware/
Host Name:	grace.hprc.tamu.edu	
Operating System:	Linux (CentOS 7)	
Total Compute Cores/Nodes:	45,376 cores 940 nodes	
Compute Nodes:	800 48-core compute nodes, each with 384GB RAM 100 48-core GPU nodes, each with two A100 40GB GPUs and 384GB RAM 9 48-core GPU nodes, each with two Quadro RTX 6000 24GB GPUs and 384GB RAM 8 48-core GPU nodes, each with 4 T4 16GB GPUs 15 48-core GPU nodes, each with two A40 48GB GPUs and 384GB RAM 8 80-core large memory nodes, each with 3TB RAM	
Interconnect:	Mellanox HDR 100 InfiniBand	
Peak Performance:	6.3 PFLOPS	
Global Disk:	5PB (usable) via DDN Lustre appliances for general use 1.4PB (usable) via Lenovo DSS GPFS appliance (purchased by and dedicated for Dr. Junjie Zhang's CryoEM Lab) 1.9PB (usable) via Lenovo DSS GPFS appliance (purchased by and dedicated for Dr. Ping Chang's iHESP Lab)	
File System:	Lustre and GPFS	
Batch Facility:	Slurm by SchedMD	
Location:	West Campus Data Center	
Production Date:	Spring 2021	

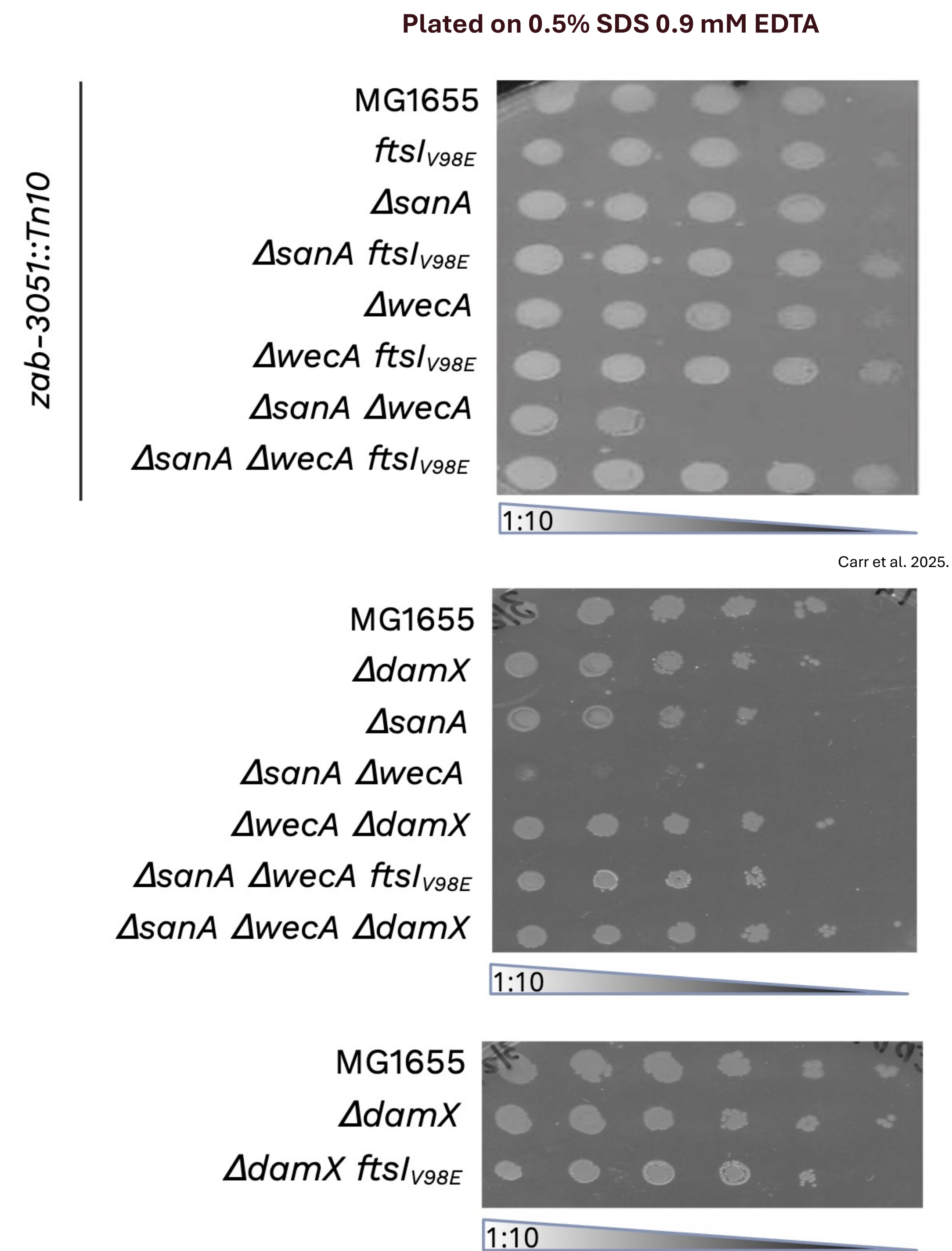
AlphaPulldown: Automated AlphaFold-based screening for SanA interaction partners



- AlphaFold enables high-accuracy structural modeling of protein complexes and can be used to predict potential protein-protein interactions.
- AlphaPulldown automates large-scale PPI with AlphaFold, allowing efficient and systematic prediction of interaction partners.

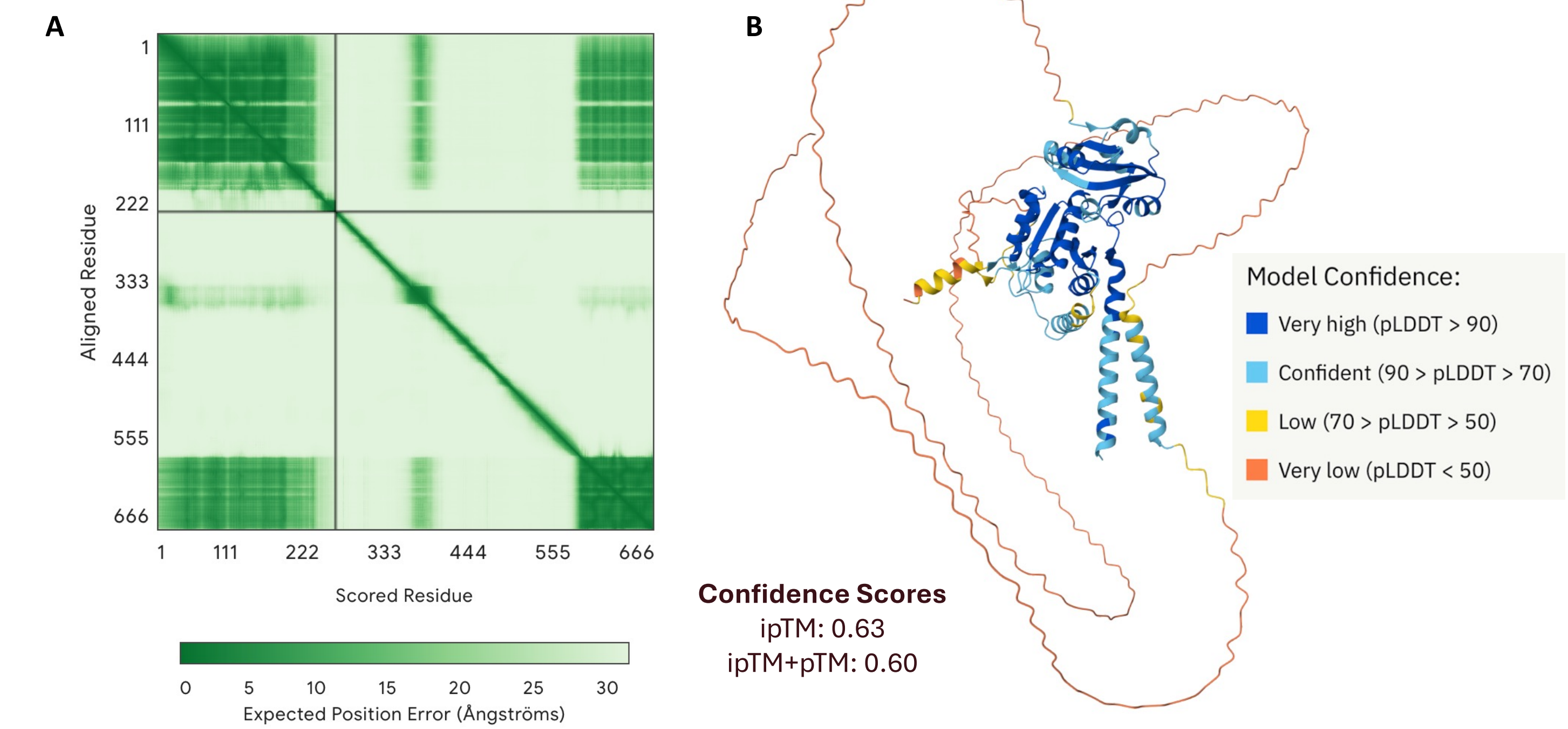
damX Suppresses a *sanA wecA* Mutation

- EOP (Efficiency of Plating): measures how well strains grow in the presence of SDS/EDTA.
- Lower EOP: growth disappears earlier across the serial dilutions → indicates sensitivity.
- Higher EOP: growth persists across more dilutions → indicates resistance.



DamX Emerges as a Possible SanA Interaction Partner

- PAE: confidence in the relative positioning of proteins in the complex.
- ipTM: overall confidence that the proteins interact.
- ipTM+ptm: confidence in the protein structure.



(A) Predicted Aligned Error (PAE) plot for the SanA-DamX complex. (B) Predicted 3D structure of the SanA-DamX interaction.

Future directions

- Crosslinking + Pull-Down: Experimentally test whether SanA and DamX interact in *E. coli*.
- Validate Interaction: Follow with western blots to confirm SanA-DamX co-enrichment.
- Expand to Other SPOR Proteins:
 - FtsN: Express from plasmid; add His/FLAG tag via Gibson.
 - DedD: Available in Keio and ASKA collections.
 - RlpA: Available in Keio and ASKA collections.
- Phenotypic Testing: Repeat EOP assays for additional SPOR proteins to identify shared envelope phenotypes.

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